SignSaathi: An Accessible, Multilingual Companion for City Signage

Megathon 2026 · Accessible AI for All · Use Case 3: Navigating City Signage · Team: **Inference Lab** · M1 Approach Document

1. Abstract

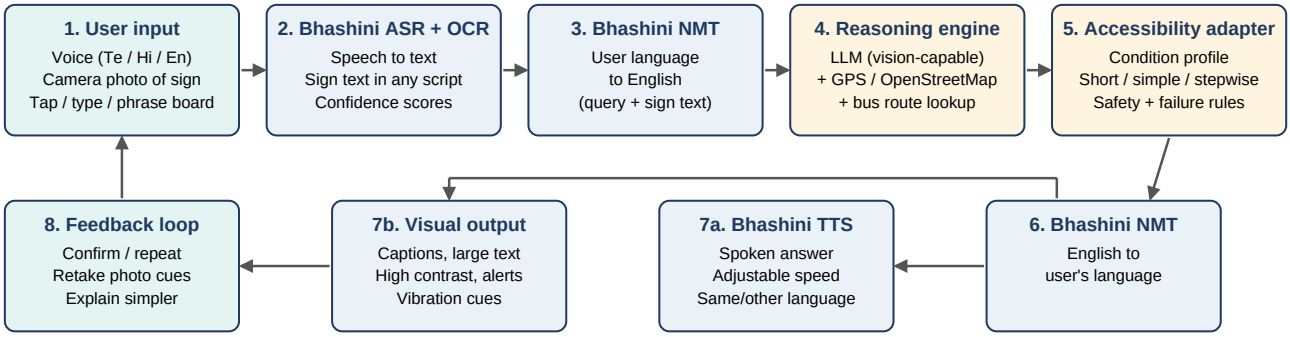
Street signs, bus boards, shop boards and pedestrian signals in Indian cities are visual, often multi-script, and rarely accessible. General AI assistants can read a photo, but they assume the user can see the screen, read English, and consume long text answers. **SignSaathi** is an accessibility layer that sits between the user and a general-purpose AI assistant. A user points their phone at a sign and asks a question in their own language (English, Hindi or Telugu); the system reads the sign, reasons about the question, and replies with a short, task-focused answer delivered in the form that suits the user's condition: speech for blind users, captions and visual alerts for deaf users, and slow, stepwise, large-text guidance for older adults. Every exchange runs through the Bhashini speech-to-speech pipeline, and the app runs on a normal Android phone or laptop with no special hardware.

2. Target Users and Chosen Conditions

Condition	Barrier at a street / bus stop	Specific feature we will demonstrate
Visual (mandatory)	Cannot see signs, screens or framing of the photo	Fully voice-first flow; camera framing cues ("tilt up, the board is cut off"); answers with distance and direction ("bus stop, 30 m ahead on your right")
Hearing	Cannot hear voice replies or bus/station announcements	Live captions of all replies; announcement capture, transcription and translation; vibration and flash alerts when the requested bus is mentioned
Ageing	Small text, complex steps, low tech comfort	Large text and high contrast, slower TTS, one instruction at a time, a single "Explain simpler" button and "Repeat" button

Condition inference: the system watches interaction signals (voice-only use with no screen taps suggests visual; typing or muted audio suggests hearing; repeated replays, zooming or long pauses suggest ageing), proposes a profile, and confirms it with the user ("Shall I speak slower?"). The user can always override it.

3. Approach and Proposed Architecture



Pipeline: ASR and/or OCR, then Translation, then AI, then Translation, then TTS, with all language steps on Bhashini. The reasoning engine receives the translated question, the extracted sign text and (when needed) the photo, GPS position and map data, and returns a structured answer. The **accessibility adapter** is the core of our work: it rewrites the raw answer to match the user's profile before translation and output.

Key scenarios covered:

- **Signboards and shop boards:** read in the user's language; answer "what is this shop?" and "is it open?" from timings on the board.
- **Bus boards:** answer only the question asked ("Bus 219 goes to the station; it is the third route listed") instead of reading the whole board.
- **Traffic signs and pedestrian signals:** describe what the signal shows, always followed by a caution; the system never declares that it is safe to cross.
- **Guidance and emergencies:** short spoken steps to a nearby landmark using GPS + OpenStreetMap; priority reading of exit, platform and emergency signs.

4. Accessible Outputs (not just an accessible interface)

- **Answer-first, then offer more:** one sentence that answers the question, followed by an offer ("Want the next two buses?") rather than a dense list or table.
- **Structure in words:** tables, timetables and route maps are converted to spoken summaries; on screen we use real headings, labelled buttons and lists for screen readers (TalkBack).
- **User controls:** Explain simpler, Repeat, adjustable speech speed and text size, and a switch to answer in a different language from the one spoken.
- **Raw vs accessible view:** a side-by-side mode that shows the plain assistant reply next to the SignSaathi reply, used in testing and in the demo.

5. Graceful Failure and Safety

- **Unclear photo or low OCR confidence:** say what was read, state what is missing, and guide a retake ("Move the phone left and hold still"). The system never guesses a route silently.
- **Misrecognised speech:** repeat back the understood question and ask for confirmation before acting; offer a phrase board as a fallback.
- **Network drop:** notify the user in their chosen mode, retry, and fall back to on-device OCR (Google ML Kit) to read sign text offline, with cached TTS prompts.
- **Safety boundaries:** signal readings are informational only; no medical, legal or financial advice is given.

6. Planned Technology (all external resources will be declared)

Layer	Planned choice
Frontend	Progressive web app usable on Android Chrome and laptop; camera, microphone, vibration and geolocation APIs; TalkBack-compatible markup
Backend	Python (FastAPI) orchestration service handling the pipeline, condition profile and failure logic
Language	Bhashini Sandbox APIs: ASR, NMT and TTS for English, Hindi and Telugu; Bhashini OCR where available
Reasoning	A vision-capable LLM via API (prompted with an accessibility rubric); OpenStreetMap for landmarks and walking directions
Offline fallback	Google ML Kit on-device text recognition for basic sign reading without network

7. Testing Methodology

Test type	Method	What we measure
Component accuracy	Test set of about 50 photos of real Hyderabad signs (bus boards, shop boards, traffic signs) in Telugu, Hindi and English, plus recorded spoken queries	OCR text accuracy; ASR word error rate per language; translation quality via native-speaker rating (1 to 5)
End-to-end tasks	Scripted tasks per condition, e.g. "find which bus goes to Secunderabad station" and "is this pharmacy open?"	Task success rate, number of turns, time to answer, end-to-end latency
Accessibility	Complete each task with eyes closed using TalkBack only; with audio muted; and at large text with slow speech; automated checks with Lighthouse / axe	Tasks completed without sight or sound; WCAG issues found and fixed
Raw vs accessible	Same question asked to a plain AI assistant and to SignSaathi; blind ratings by team members and volunteers	Answer length, time to the key fact, clarity rating
Robustness	Blurred, tilted and partly hidden sign photos; background street noise; throttled and dropped network	Correct retake / repeat prompts; zero silent guesses; offline fallback works
User validation (bonus)	Short sessions with visually impaired users or older adults if available during the event	Qualitative feedback and observed difficulties, reported honestly with limitations

8. Scope, Limitations and Build Plan

- **In scope:** English, Hindi and Telugu input and output; the three conditions above; sign reading, bus-board questions, signal descriptions and short walking guidance.
- **Out of scope:** training our own LLM, special hardware, and real-time obstacle or vehicle detection; the app supports, and does not replace, a white cane or human help.
- **Known limitations:** OCR accuracy drops on worn, handwritten or night-time signs; GPS drift of several metres in dense areas; LLM and Bhashini latency over mobile data.
- **Build plan:** (1) Bhashini speech-to-speech loop live; (2) sign OCR + reasoning for bus boards and shop boards; (3) accessibility adapter and the three condition modes; (4) failure handling and offline fallback; (5) testing as above, demo video and write-up.